

LitSense Weekly Progress Report

November 24, 2025

Focus: Implemented intelligence core with multi-agent infrastructure

Achievement: 7 agents operational, real-time collaboration working, 3 modules complete

Module Progress Status

#	Module Name	Status	%
2	Interaction Analysis	Complete	100%
3	Discussion Analysis	In Progress	90%
4	Role Assignment	Complete	100%
5	Matching System	Complete	100%
1,6-10	Remaining Modules	Pending	0%

Multi-Agent Infrastructure

Agent Implementation

- Agent 1:** Understanding Detection - monitors every 5s, detects struggles
- Agent 2:** Peer Matching - finds helpers, scores 0-100
- Agent 3:** Discussion Facilitation - manages conversations
- Agent 4:** Annotation Analysis - pattern recognition
- Agent 5:** Journey Logging - tracks learning progression
- Agent 6:** AI Explanations - GPT-4 tutoring
- Agent 7:** Smart Notifications - context-aware alerts

AI Coordination Core

- Centralized event routing system
- Prevents agent conflicts
- Maintains global state
- Unified query interface

Data Collection

- Eye tracking via WebGazer.js
- Highlight patterns and frequency
- Time spent per section
- Understanding scores (0-100)
- Collaboration events log

Agent Functionality Details

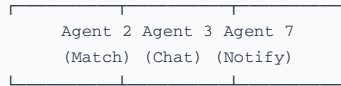
Agent	Input Data	Output/Action
Agent 1	Eye tracking, highlights, time data	Understanding score, struggle events
Agent 2	User profiles, section data	Peer matches with scores
Agent 3	Confused highlights, user context	Chat channels, conversation starters
Agent 4	Annotation patterns, highlights	Topic analysis, collaboration suggestions
Agent 5	All user events, timestamps	Journey logs, session summaries
Agent 6	Confused text, user questions	AI explanations, examples
Agent 7	Struggle events, peer availability	Targeted notifications

System Architecture

Researcher Interaction → Interaction Collector → Agent 1



AI Coordination Core (Event Router)

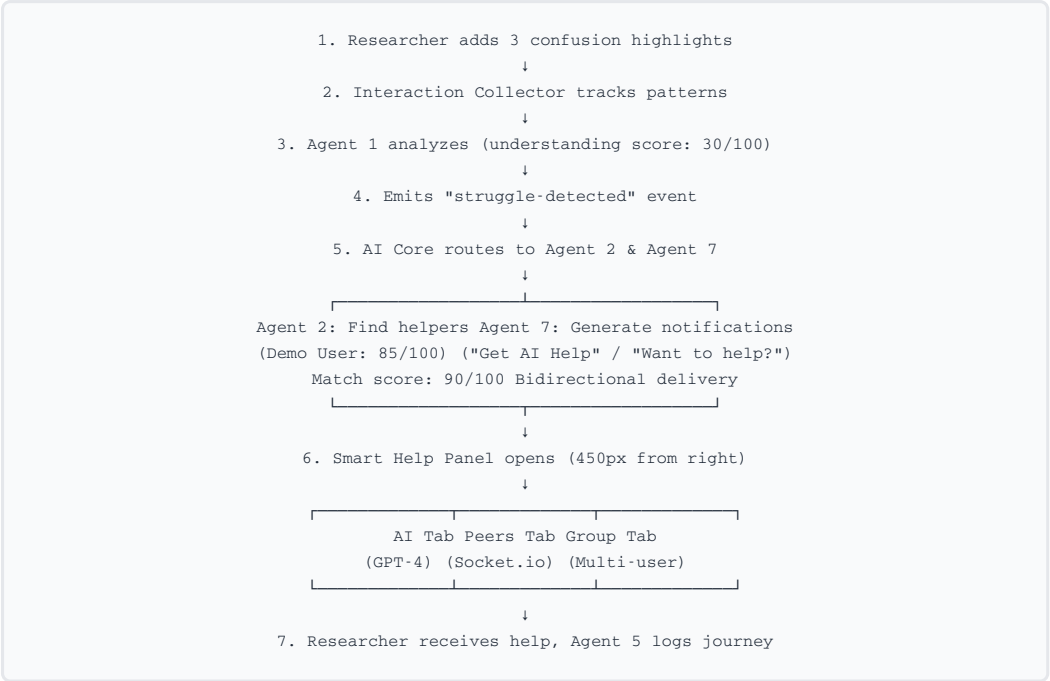


Smart Help Panel → User gets help

System Workflow & Implementation

November 24, 2025

Complete System Workflow



Data Collection System

Data Type	Collection Method	Usage
Eye Tracking	WebGazer.js (every 100ms)	Attention patterns, confusion detection
Highlights	User annotations with reason	Understanding analysis, peer matching
Time Spent	Section entry/exit timestamps	Struggle detection, engagement metrics
User Actions	Click events, scrolling, revisits	Behavioral pattern analysis
Collaboration	Chat messages, help requests	Effectiveness tracking, journey logging

Real-Time Collaboration Infrastructure

Implementation

- Socket.io server on port 3000
- Document-based room architecture
- Instant highlight synchronization
- User presence tracking
- Message latency under 100ms
- Automatic reconnection handling

Features Working

- Real-time highlight broadcast
- Section assignment sync
- User join/leave notifications
- Bidirectional notifications
- Peer status updates
- Chat message delivery

AI Integration

Component	Implementation	Purpose
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AI Tutor Endpoint	/api/ai-tutor using OpenAI GPT-4	Quick explanations for confused highlights
Context Injection	Automatic highlight text inclusion	Relevant, focused responses
Conversation History	Maintained via API state	Coherent multi-turn conversations
Educational Prompts	Custom system prompts for tutoring	Simplified, encouraging explanations

Problems Solved This Week

Challenge	Solution Implemented	Status
Notifications to all users	Added strict targetUserId filtering at event level	90% Fixed
User list inconsistency	Unified Socket.io server, eliminated polling	Resolved
Agent coordination conflicts	Built AI Coordination Core with event routing	Resolved
Agent 2 finding 0 peers	Auto-registration + section tracking logic	Resolved

Next Two Weeks Plan

Period	Planned Implementation
Week 1	Module 1: Reflection Analysis - paragraph highlighting, comment quality tracking Module 6: Fact-checking - API integration for claim verification Module 3: Complete peer chat Socket.io integration
Week 2	Module 7: Related Work Analysis - paper similarity detection Module 8: Summary Builder - automated section summaries